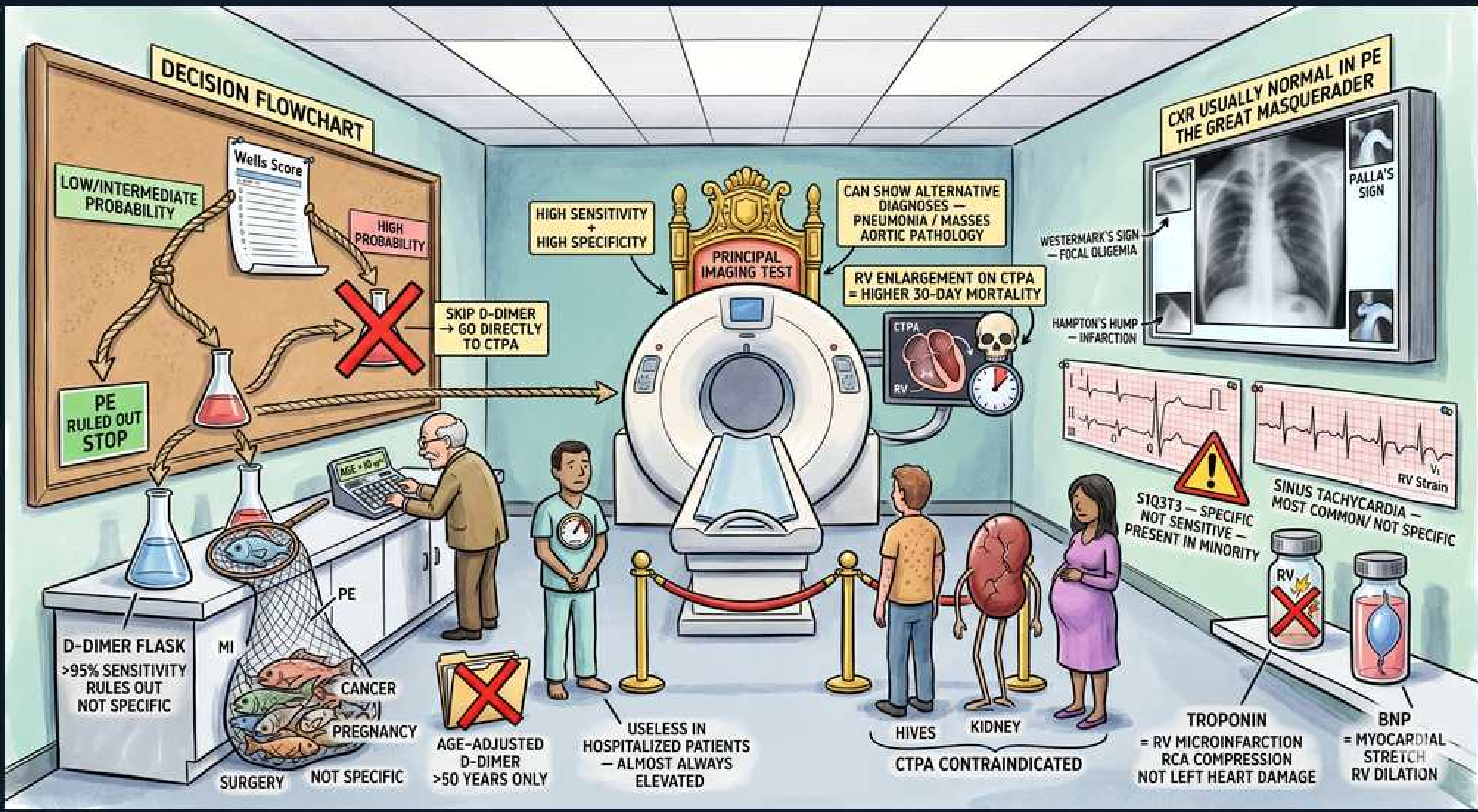
VTE — PE Diagnosis: Wells, D-dimer, CTPA & Biomarkers



1. Wells score decision flowchart

WHAT YOU SEE
Decision flowchart / Wells score board

WHAT IT ENCODES
The Wells criteria stratify pretest probability for PE. Low/intermediate probability → D-dimer first; high probability → skip D-dimer and go directly to CTPA. This gates resource use and prevents over-imaging.

BOARD TRAP
High pretest probability → go STRAIGHT to CTPA; a negative D-dimer here does NOT rule out PE.

2. D-dimer flask — sensitive not specific

WHAT YOU SEE
D-dimer flask, '>95% sensitivity rules out, not specific'

WHAT IT ENCODES
D-dimer has high sensitivity (>95%) but low specificity. A negative result in low/intermediate probability reliably EXCLUDES PE. It is a rule-OUT, never a rule-IN test.

BOARD TRAP
A POSITIVE D-dimer does not confirm PE — it is nonspecific; only a NEGATIVE result (in low pretest probability) is useful to exclude.

3. False-positive D-dimer causes

WHAT YOU SEE
MI, cancer, pregnancy, surgery icons

WHAT IT ENCODES
D-dimer rises in MI, malignancy, pregnancy, recent surgery, sepsis, and advanced age — limiting specificity. In these settings a positive value is expected and unhelpful.

BOARD TRAP
Don't order D-dimer in a post-op or pregnant patient expecting it to rule in PE — it's predictably elevated and uninformative.

4. Age-adjusted D-dimer >50 yrs

WHAT YOU SEE
'Age-adjusted D-dimer >50 years only'

WHAT IT ENCODES
In patients >50, use an age-adjusted cutoff (age × 10 µg/L) to improve specificity without losing sensitivity, reducing unnecessary CTPA in the elderly.

BOARD TRAP
Use a FIXED cutoff in patients ≤50; age-adjustment applies only above 50 years.

5. Skip D-dimer → CTPA

WHAT YOU SEE
'Skip D-dimer, go directly to CTPA' arrow

WHAT IT ENCODES
With high pretest probability the post-test probability after a negative D-dimer is still too high to exclude PE, so proceed directly to definitive imaging (CTPA).

BOARD TRAP
D-dimer is a screening filter for LOW/INTERMEDIATE risk only — bypass it when probability is high.

6. CTPA — principal imaging test

WHAT YOU SEE
CT scanner labeled 'principal imaging test'

WHAT IT ENCODES
CT pulmonary angiography is the gold-standard confirmatory test for PE. It also detects alternative diagnoses (pneumonia, masses, aortic dissection) and shows RV enlargement, which predicts higher 30-day mortality.

BOARD TRAP
RV enlargement on CTPA (RV:LV >1) is a prognostic red flag for early mortality — report it, don't ignore it.

7. CTPA contraindicated

WHAT YOU SEE
Pregnant patient, kidney, hives icons

WHAT IT ENCODES
CTPA is contraindicated/limited by contrast allergy, renal impairment, and relative concerns in pregnancy. In these cases use V/Q scanning or, for DVT, compression ultrasound.

BOARD TRAP
Contrast allergy or AKI → V/Q scan, not CTPA. In pregnancy, leg ultrasound first; V/Q if needed.

8. CXR normal — great masquerader

WHAT YOU SEE
Chest X-ray, 'CXR usually normal in PE'

WHAT IT ENCODES
The chest X-ray is usually normal in PE (the 'great masquerader'); its main role is to exclude other causes. Classic but insensitive signs: Westermark sign (focal oligemia), Hampton's hump (pleural-based infarct), Pallas sign.

BOARD TRAP
A normal CXR does NOT exclude PE — it argues against pneumonia/effusion as alternatives, nothing more.

9. Westermark / Hampton's hump

WHAT YOU SEE
Focal oligemia and wedge infarct on CXR

WHAT IT ENCODES
Westermark sign = regional oligemia distal to occlusion; Hampton's hump = wedge-shaped pleural-based opacity from infarction. Both are specific but rare and insensitive.

BOARD TRAP
These signs are nearly pathognomonic but seen in a minority — their ABSENCE never rules PE out.

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10. ECG — S1Q3T3

WHAT YOU SEE
'S1Q3T3 specific not sensitive'

WHAT IT ENCODES
The S1Q3T3 pattern (S in I, Q and inverted T in III) reflects acute RV strain; it is specific but present in a minority. Sinus tachycardia is the MOST COMMON ECG finding in PE.

BOARD TRAP
Most common ECG in PE is sinus TACHYCARDIA, not S1Q3T3 — S1Q3T3 is classic but uncommon.

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11. Sinus tachycardia — most common

WHAT YOU SEE
'Sinus tachycardia most common'

WHAT IT ENCODES
Sinus tachycardia is the single most frequent ECG abnormality in PE, reflecting hypoxia and sympathetic drive. New RBBB or right-axis deviation also suggest RV strain.

BOARD TRAP
Don't pick S1Q3T3 as 'most common' — that's a distractor; the answer is sinus tachycardia.

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12. Troponin — RV microinfarction

WHAT YOU SEE
Troponin vial, 'RV microinfarction'

WHAT IT ENCODES
Troponin elevation in PE reflects RV microinfarction from strain, not left coronary disease; it marks intermediate-risk PE and worse prognosis.

BOARD TRAP
Troponin+ in PE ≠ ACS — it's RV strain and a risk-stratifier, not an indication for coronary angiography.

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13. BNP — RV dilation/stretch

WHAT YOU SEE
BNP vial, 'myocardial stretch / RV dilation'

WHAT IT ENCODES
BNP/NT-proBNP rise from RV myocardial stretch in PE and identify intermediate-risk patients. Combined with troponin and RV imaging, they refine prognosis and disposition.

BOARD TRAP
Elevated BNP in PE signals RV strain/intermediate risk — it does not mean coexisting LEFT heart failure.

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14. Pretest probability gating

WHAT YOU SEE
Arrows linking Wells → imaging path

WHAT IT ENCODES
The whole algorithm is probability-driven: low/intermediate → D-dimer → CTPA if positive; high → CTPA directly. PERC rule can avoid even D-dimer in very low-risk patients.

BOARD TRAP
Skipping the pretest probability step and ordering D-dimer on everyone causes false-positive cascades — always risk-stratify first.
